

DATA ANALYTICS CASE STUDY

Automated Banking Analytics Dashboard

SQL Server • Microsoft Fabric • Power BI

End-to-end retail banking intelligence, from raw transactions to business decisions

Business Problem Statement

1

No unified view

Transactions, branches, and customers analyzed in silos — no single source of truth.

2

Manual reporting

MIS reports built manually in spreadsheets, taking hours to days per request.

3

Inconsistent KPIs

Success Rate and other metrics defined differently across teams, causing mismatched numbers.

4

Reactive, not proactive

Failures, underperforming branches, and cross-sell gaps are found late, after the impact has landed.



Questions like ‘which channel is failing most’ or ‘which customers hold only one account’ took hours of manual SQL work to answer — by the time an answer arrived, the decision window had passed.

— Core motivation for this project

Business Objectives

1

Automate the pipeline

Ingest, clean, and refresh transaction data daily with zero manual touch.

2

Unify KPI definitions

One governed semantic model so every team sees the same numbers.

3

Surface risk early

Flag failing channels and underperforming branches before losses grow.

4

Enable cross-sell

Segment and score customers by value, income, and product depth.

5

Decision-ready output

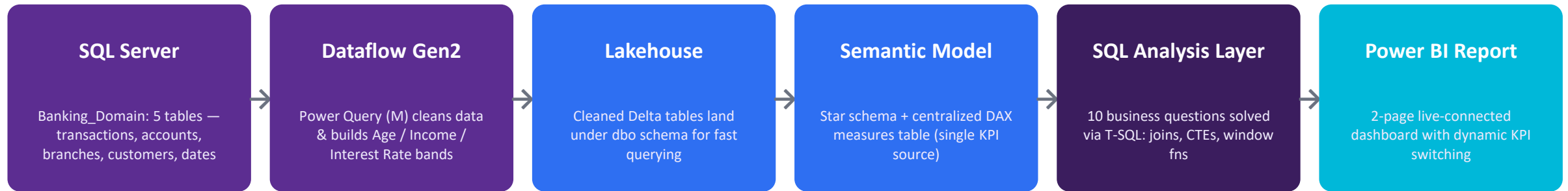
Convert raw data into ranked, actionable recommendations — not just charts.

6

Build for scale

A star-schema foundation that extends to new products and channels.

End-to-End Architecture



Two-track delivery

Analytical track: 10 SQL Server queries run directly against Banking_Domain, each mapped to one business question and one operational recommendation.

Visual track: Fabric pipeline refreshes the Power BI dashboard daily, at 10:00 PM, with no manual trigger — building on the same governed semantic model the SQL layer queries.

Tools & Techniques Used

Data Platform

- SQL Server (Banking_Domain)
- Microsoft Fabric Lakehouse
- On-Premises Data Gateway
- Fabric Data Pipeline (scheduled refresh)

Data Prep & Modelling

- Dataflow Gen2 / Power Query (M)
- Star schema design
- DAX measures & field parameters
- Dynamic KPI + dynamic title logic

SQL Analysis Techniques

- GROUP BY + JOIN aggregation
- CTEs + CASE scoring logic
- Conditional aggregation
- Window functions (LAG), multi-table joins

Key Insights from SQL Analysis (1/2)

1

Mass segment drives volume

Mass customers generate ₹89.7 Cr in transaction value (4,600 customers) — more than Affluent + Premium combined, despite lower per-customer average.

2

POS & Internet Banking fail most

POS (4.05%) and Internet Banking (4.03%) post the highest failure rates; Mobile Banking carries the largest revenue-at-risk at ₹1.58 Cr.

3

Savings accounts, biggest value

Savings accounts drive ₹67.5 Cr in transaction value — the largest of all 5 account types — while Loan accounts carry the highest interest rate (10.99%).

4

3–8L income band is the core base

Customers earning ₹3–8 lakhs generate ₹75.3 Cr in value across ~5,000 customers — the single largest income segment by volume and value.

Key Insights from SQL Analysis (2/2)

5

Transaction volume is seasonal

Monthly counts swing from a February low (3,909) to an October peak (4,302) — a ~10% seasonal range worth planning staffing/capacity around.

6

Failure risk concentrates by role

Premium-segment Engineers show the highest failure rate at 4.73%, ahead of every other segment + occupation combination tested.

7

Clear cross-sell shortlist exists

High-income (>₹8L) customers holding only one account are flagged as top cross-sell priority — an immediately actionable target list.

8

3 branches need priority action

A composite score (revenue-at-risk + failure rate + volume) ranks Jaipur Central (East), Jaipur Central (North), and Chennai Central (South) as this quarter's top intervention targets.

Live Dashboard Snapshot



Executive Overview page: dynamic KPI switching, monthly trend, channel mix, and regional breakdown — all filterable by City, Region, and Year.

Customer Insights page: segment, occupation, income band, and age band views — built for cross-sell and demographic targeting.

Design standard

- One consistent purple / royal-blue brand palette across both pages
- Donut-center callouts, sorted bars, and dynamic chart titles
- Shared City/Region filters and Key Metrics selector on every page
- Live Connection to the Fabric semantic model — always current

Business Recommendations

1

Fix POS & Internet Banking reliability first

These two channels post the highest failure rates — route an IT/ops review here before other initiatives.

2

Launch a cross-sell campaign

Target high-income, single-account customers surfaced by the SQL analysis — a ready-made lead list, no extra modelling needed.

3

Prioritize 3 branches this quarter

Jaipur Central (East & North) and Chennai Central (South) score highest on the composite risk index — focus operational support there.

4

Plan capacity around seasonality

Build October-level staffing/infra headroom given the ~10% swing between February lows and October peaks.

5

Retain the Mass segment

It drives the largest share of transaction value — protect it with service-quality monitoring, not just premium-tier focus.

Thank You

Automated Banking Analytics Dashboard — SQL Server, Microsoft Fabric & Power BI

1

GitHub Repository

<https://github.com/seema-kri/Automated-Banking-Analytics-Dashboard>

2

Live Dashboard (Power BI)

https://app.fabric.microsoft.com/links/gewlC3bykT?ctid=e93d71d6-b5c0-4b78-a861-d9964ecdcd6&pbi_source=linkShare&bookmarkGuid=0bf56dac-f8bf-486a-abbb-a46c2251f503

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Business Requirements Document

Included in /docs folder of the repository